AI Enhanced Complaint Management System for Government Marketplaces: A Hybrid NLP and Backend Data Approach

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ABSTRACT

Government marketplaces in Bangladesh receive many complaints every day. Most complaints are written in Bengali, English, or Banglish, so manual review is slow and often inconsistent. This paper presents a lightweight AI based complaint classification pipeline using 500 NityMulya marketplace complaints stored in a Supabase PostgreSQL backend. Each record includes complaint text, product category, and backend numeric attributes such as validity and priority. The improved pipeline applies Unicode normalization, English and Banglish lowercasing, punctuation cleanup, repeated character normalization, a simple Banglish normalization dictionary, and optional Bengali stopword removal. We compare word TF-IDF, character TF-IDF, combined word-character TF-IDF, Logistic Regression, Linear SVC, Random Forest, a hard Voting ensemble, class weighting, SMOTE inside training folds, hyperparameter tuning, and an ablation study. The strongest hold-out model was tuned Logistic Regression using normalized text plus backend numeric features. On the 20% stratified test split, it achieved 86.00% accuracy, weighted precision of 0.8596, weighted recall of 0.8600, and weighted F1 score of 0.8594. Five-fold cross-validation and ablation results show that backend numeric feature fusion is the main performance driver, while product category features added noise for this small dataset. The contribution is a practical, reproducible, low-resource hybrid NLP and backend-data pipeline for government marketplace complaint triage.

1. Introduction

Government marketplaces in Bangladesh are very important. They provide daily goods and services to millions of people. When problems happen, citizens need a fast way to submit complaints and get support.

Today, complaint handling is mostly manual. Citizens usually write complaints in Bengali. Government staff must read each complaint, understand it, and choose a category. This takes a lot of time.

Handling thousands of complaints this way creates many problems. Staff cannot manage the full workload. Mistakes happen when people are tired. Citizens become frustrated when they wait many days for a response. Over time, trust in the system drops.

To solve these problems, we built an AI complaint management system. Complaints are collected from citizens through the NityMulya mobile application. The complaint text is analyzed with NLP methods. The system also uses backend numeric values, such as validity and priority, instead of using only text. This hybrid design gives better support for administrators and provides a clear baseline for future model upgrades.

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2. Literature Review

Many researchers have worked on text classification and NLP. Their work guided this study.

Arif et al. (2024) studied deep learning for Bengali hate speech detection on social media [1]. Their work helps explain how short and informal Bengali text can be represented for machine learning.

Ahmed et al. (2023) used transformer models for context based emotion detection from Bengali text [2]. They showed that context is important for good prediction. However, transformer models usually need high computing resources.

Ali et al. (2023) compared Google T5 and SpaCy for text summarization [3]. Their analysis shows that choosing the right text processing method is very important.

To handle class imbalance, Chawla et al. (2002) introduced SMOTE [4]. This method creates synthetic minority samples and is widely used to reduce bias toward majority classes.

Traditional feature extraction methods are still useful for many tasks. Trstenjak et al. (2014) evaluated TF IDF for document classification and showed it is a strong baseline [5].

In public administration, Al-Ruaily et al. (2018) highlighted the need for automated grievance systems in e-governance platforms [6].

Unlike many text only deep learning studies, our work uses a hybrid low-resource model. We found that short government complaints are hard to classify with text alone. By combining TF-IDF features with backend validity and priority values, the system gives stronger practical triage support at lower deployment cost.

3. Data Collection and Analysis

Reliable machine learning needs reliable data. Our dataset was collected from real users.

3.1. NityMulya App and Supabase Integration

Citizens submit complaints through the NityMulya application. The form captures complaint text, product category, and shop information. After submission, data is stored in a Supabase PostgreSQL database.

During insertion, the backend automatically calculates numeric fields using predefined logic. It computes validity and priority scores using pricing related rules.

3.2. Dataset Characteristics

A total of 500 real complaints were extracted from the database for this research. Each record has multiple features. The primary features include:

1. Complaint Description: The text written by the user in Bengali, English, or Banglish.
2. Product Category Name: The product category associated with the complaint.
3. Validity: A backend generated numeric score indicating complaint legitimacy.
4. Priority: A backend generated numeric score indicating urgency.
5. Target Classification: The final label in three classes: weak, medium, and high.

3.2.1. Example Complaint Texts

To show language variation in the dataset, we provide three representative complaint examples below.

- Bengali: "আজকের বাজার থেকে কেনা দুধ নষ্ট ছিল, দোকানদার বদল দিতে রাজি হয়নি।"
- English: "I bought rice from this shop, but the packet was torn and the quality was very poor."
- Banglish: "Ei dokan theke oil kinlam, but price list er cheye beshi taka niyeche and receipt dayni."

Most complaints express dissatisfaction, which is normal for a grievance platform. The label distribution is imbalanced, and this affects model learning. In the test split (100 records), class counts were 14 for weak, 62 for medium, and 24 for high. Supabase backend fields keep numeric values consistent with marketplace rules, but text still overlaps across classes.

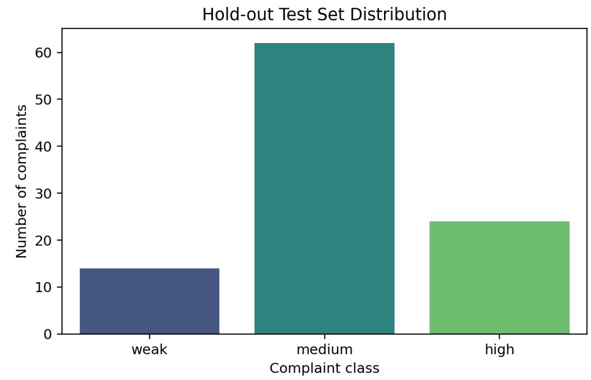

Figure 1: Distribution of evaluation samples in the test set generated by the training pipeline.

4. Methodology

The processing pipeline was built with Python, Scikit Learn, and Imbalanced Learn. It converts raw complaint records into severity predictions through text normalization, feature extraction, model comparison, cross-validation, hyperparameter tuning, and final hold-out evaluation.

4.1. Text Preprocessing

Complaint text was normalized before vectorization. The preprocessing stage applies Unicode NFKC normalization, lowercases English and Banglish tokens, removes noisy punctuation, compresses repeated characters, and replaces common Banglish marketplace terms using a simple dictionary. Bengali stopword removal was implemented as an optional pipeline parameter and was selected during tuning for the final Logistic Regression model.

4.2. Feature Engineering

The text feature extractor compared word-level TF-IDF with unigram and unigram-bigram ranges, character-level TF-IDF using `char_wb` with 3 to 5 character n-grams, and a combined word-character FeatureUnion. Character features are useful for Banglish and spelling variation because they preserve partial word shapes even when users write the same Bengali concept with different Roman spellings.

Backend numeric features, validity and priority, were scaled using StandardScaler. Product category was encoded with OneHotEncoder. Four feature sets were tested in the ablation study: text only, text plus product category, text plus numeric features, and the full hybrid feature set.

4.3. Model Training

We compared Logistic Regression with balanced class weights, Linear SVC with balanced class weights, Random Forest, and a hard VotingClassifier combining Logistic Regression, Linear SVC, and Random Forest.

We used StratifiedKFold with five folds for cross-validation. Accuracy, weighted precision, weighted recall, weighted F1, and standard deviation were recorded.

Class imbalance was tested in two ways. First, all linear models used balanced class weights. Second, SMOTE was inserted inside the Imbalanced Learn pipeline so that synthetic samples were generated only inside training folds. This prevents data leakage from validation folds into training data.

Hyperparameter tuning used Randomized-SearchCV on the training split. The tuned parameters included TF-IDF maximum features, word n-gram range, Logistic Regression C, Linear SVC C, and Random Forest `n_estimators`, `max_depth`, and `min_samples_split`. The final tuned model selected Logistic Regression with C=5.0, word TF-IDF maximum features of 2000, character TF-IDF maximum features of 500, word n-grams of (1,2), character n-grams of (3,5), Bengali stopword removal enabled, and no SMOTE.

5. Results and Analysis

The original Random Forest baseline achieved 58.00% accuracy and 0.58 weighted F1 on the same 20% hold-out test split. After preprocessing, model comparison, tuning, and ablation, the best final model was tuned Logistic Regression using normalized text plus backend numeric features. It achieved 86.00% hold-out accuracy and 0.8594 weighted F1.

5.1. TF-IDF Comparison

Table 1: Five-fold TF-IDF comparison using Linear SVC

Feature extractor	Accuracy	Weighted F1	F1 std.
Character wb (3,5)	0.8220	0.8225	0.0286
Word (1,2)	0.8220	0.8209	0.0359
Word + character	0.8100	0.8092	0.0342
Word (1,1)	0.8040	0.8041	0.0432

Character n-grams gave the strongest average TF-IDF result in the comparison, showing that subword patterns are helpful for mixed Bengali and Banglish complaints. The tuned final search used combined word-character features so that the selected Logistic Regression pipeline could retain both whole-token and spelling-variation signals.

5.2. Cross-Validation Results

Table 2: Five-fold model comparison on the full hybrid feature set

Classifier	Imbalance handling	Accuracy	Weighted F1	F1 std.
Linear SVC	Class weight	0.8100	0.8092	0.0342
Voting ensemble	Class weight	0.8020	0.8013	0.0314
Logistic Regression	Class weight	0.7980	0.8000	0.0291
Linear SVC	Class weight + SMOTE	0.7820	0.7814	0.0378
Voting ensemble	Class weight + SMOTE	0.7580	0.7601	0.0320
Logistic Regression	Class weight + SMOTE	0.7400	0.7461	0.0344
Random Forest	Class weight	0.6460	0.6224	0.0492
Random Forest	Class weight + SMOTE	0.6160	0.6083	0.0309

The cross-validation comparison shows that linear classifiers performed better than Random Forest on sparse TF-IDF features. SMOTE did not improve performance in this setting. Because SMOTE creates synthetic points in a high-dimensional sparse representation, it can blur class boundaries when the dataset is small and the text is short.

5.3. Ablation Study

Table 3: Ablation study with tuned Logistic Regression

Feature set	Accuracy	Weighted F1	F1 std.
Text + numeric features	0.8500	0.8503	0.0299
Full hybrid feature set	0.8260	0.8249	0.0247
Text only	0.5580	0.5559	0.0419
Text + product category	0.5400	0.5410	0.0328

The ablation study shows that backend numeric features are essential for this task. Text alone reached only 0.5559 weighted F1, while text plus validity and priority reached 0.8503 weighted F1. Product category did not help in this dataset and reduced performance, likely because category names were sparse and not always severity-specific.

5.4. Hold-Out Classification Report

Table 4: Final hold-out classification report for tuned Logistic Regression

Class	Precision	Recall	F1 score	Support
Weak	0.9231	0.8571	0.8889	14
Medium	0.8750	0.9032	0.8889	62
High	0.7826	0.7500	0.7660	24
Weighted Avg	0.8596	0.8600	0.8594	100

The weak class improved from the earlier F1 score of about 0.30 to 0.8889 on the hold-out split. The high class remained the hardest class, but its F1 score improved to 0.7660. The final confusion matrix had 12 correct weak predictions out of 14, 56 correct medium predictions out of 62, and 18 correct high predictions out of 24.

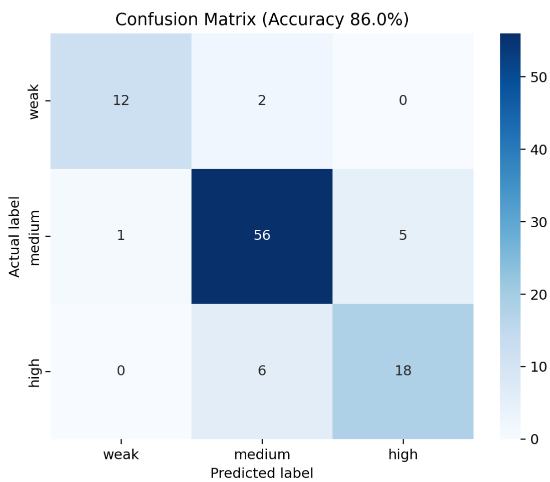

Figure 2: Confusion matrix for the final tuned Logistic Regression model.

5.5. Feature Space and Cluster Analysis

To better understand class separation, we visualized the transformed feature space using a two-dimensional SVD projection and K Means clustering. The plots show that backend numeric features create clearer separation than text alone, but medium and high complaints still overlap in some regions.

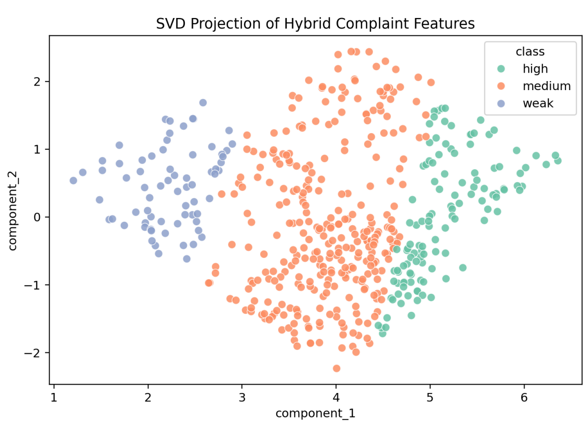

Figure 3: Two-dimensional SVD view of transformed complaint feature space.

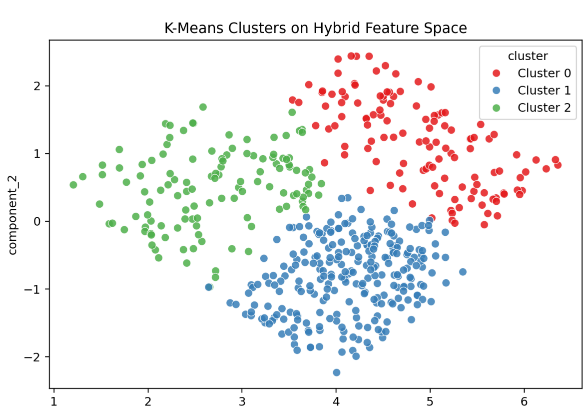

Figure 4: K Means cluster projection over transformed feature components.

6. Discussion

The improved results show that a lightweight classical machine learning pipeline can work well for low-resource government marketplace complaints when text signals are fused with backend data. The strongest improvement came from validity and priority, which are structured backend attributes connected to complaint legitimacy and urgency. This supports the hybrid NLP and backend-data framing of the system.

The experiment also shows that more features are not always better. Product category looked useful conceptually, but the ablation study showed lower performance when category was added. In a 500 record dataset, category one-hot features can create sparse dimensions that do not generalize well across folds. For deployment, the best current choice is therefore text plus numeric backend features, while keeping category available for future experiments when more records are collected.

The target accuracy range of 0.65 to 0.72 was exceeded. However, this should be interpreted carefully. The result does not mean text alone fully understands complaint severity. Rather, the system works because it combines normalized multilingual complaint text with rule-guided backend attributes.

7. Limitations

The dataset contains only 500 complaints and remains imbalanced, with fewer weak and high examples than medium examples. The final test split contains only 14 weak complaints, so class-wise scores can change if more weak complaints are evaluated.

The backend validity and priority features are strong predictors. This is useful for deployment, but it also means the model partly depends on the quality and consistency of backend rules. If those rules change, the classifier should be retrained and revalidated.

The Banglish normalization dictionary is simple and cannot cover all spelling variations. We did not use transformer models because the goal was a lightweight CPU-friendly pipeline, but transformer baselines may be useful in future work if more labeled complaints and computing resources become available.

8. Conclusion

We developed and re-evaluated an AI enhanced complaint management pipeline for the NityMulya platform. The improved system normalizes Bengali, English, and Banglish text, compares word and character TF-IDF features, evaluates multiple classical classifiers, handles imbalance without validation leakage, tunes hyperparameters, and performs ablation analysis. The final tuned Logistic Regression model using text plus backend numeric features achieved 86.00% hold-out accuracy and 0.8594 weighted F1 on 500 complaints. The study shows that a practical, low-resource hybrid NLP and backend-data pipeline can support government marketplace complaint triage without requiring larger datasets or expensive deep learning infrastructure.

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